Congratulations! You passed!

Grade received 100% **To pass** 65% or higher

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Plane Sweep: Concept

coach
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y again

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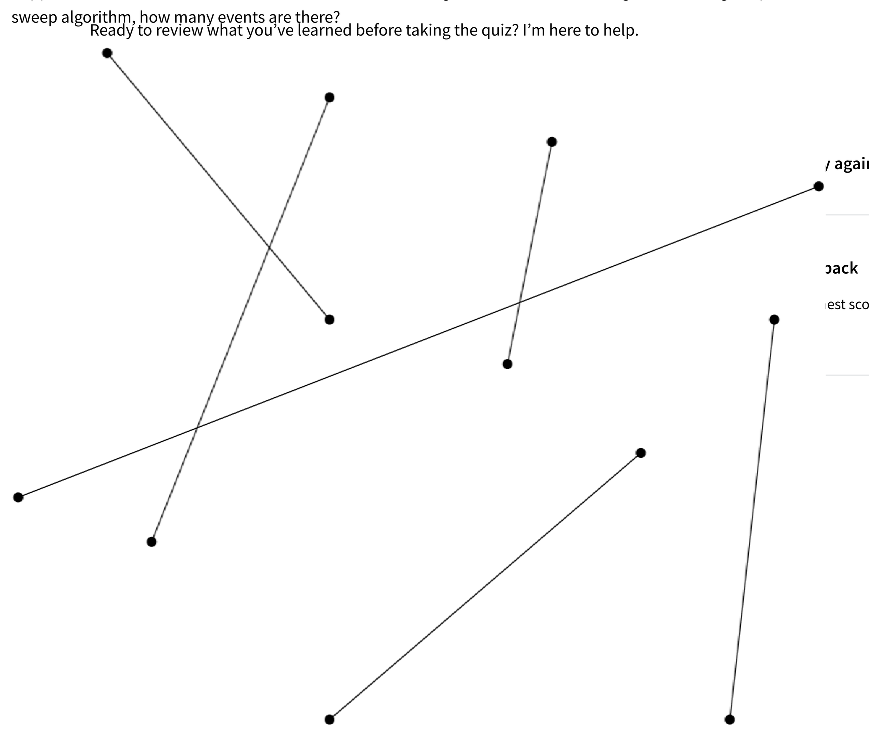
est score

1. Suppose we want to find all intersections in the set of line segments shown in the image below using the plane sweep algorithm, how many events are there?

1 / 1 point



15

Correct
Correct, there are 6 line segments and 3 intersections, which give 15 events.

2. Suppose when the plane sweep algorithm for line intersections reaches an intersection point, but the line segments are not swapped, what will happen to the output of the algorithm?

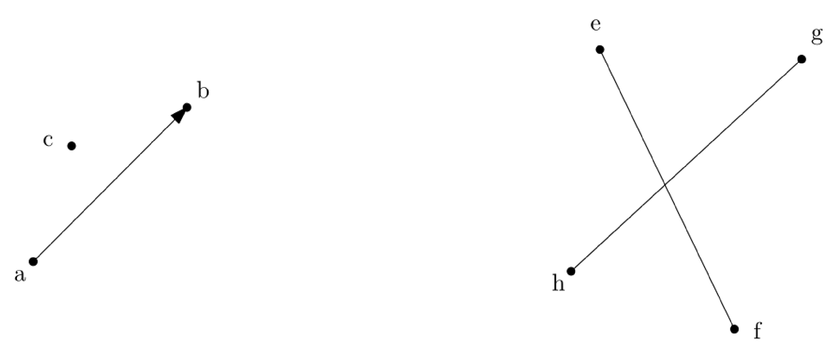
1 / 1 point

- (•) The algorithm might fail to report some of the intersections.
- ( ) The algorithm might report intersections that do not exist.
- ( ) Both of the previous options are possible.
- ( ) The algorithm might get stuck in an infinite loop and never finish.

Correct
Correct, the neighbor relation in the status structure no longer corresponds to the real situation, which causes some of the edges to not be checked for intersection.

3.

1 / 1 point



Suppose we have an orientation test **ot(a,b,c)** that takes three points as parameters. This orientation test decides whether point **c** is to the left or to the right of the directed edge from point **a** to **b**. If **c** is left, the outcome is -1. If **c** is right the outcome is 1. If **c** is on the same neither left nor right the outcome is 0. How can we use this orientation test to decide whether two line segments intersect?

- (•) ot(h,g,e) * ot(h,g,f) $\leq$ 0 and ot(f,e,h) * ot(f,e,g) $\leq$ 0
- ( ) ot(h,g,e) * ot(h,g,f) $\geq$ 0 and ot(f,e,h) * ot(f,e,g) $\leq$ 0
- ( ) ot(h,g,e) * ot(h,g,f) $\leq$ 0 and ot(f,e,h) * ot(f,e,g) $\geq$ 0
- ( ) ot(h,g,e) * ot(h,g,f) $\geq$ 0 and ot(f,e,h) * ot(f,e,g) $\geq$ 0

Correct
Correct, e and f should be on different sides with respect to edge hg, the same holds for h and g for edge ef.